Granular Origins of Agglomeration *

Shinnosuke Kikuchi
UCSD

Daniel G. O'Connor
University of Michigan

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Abstract

Large firms dominate many local labor markets. How does this granularity shape economic geography and optimal place-based policy? We develop an economic geography model with granular firms facing idiosyncratic shocks and show that average wages rise with labor market size. Using establishment-level data on Japanese manufacturing, we estimate the model and find evidence consistent with our mechanism. Granularity explains 15–37% of estimated agglomeration externalities in the smallest commuting zones, but only 2–4% in Tokyo. Optimal industrial and wage policy increases the population in the smallest cities; firm effects depend on labor market conduct.

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1 Introduction

A few large firms dominate many local labor markets. Kodak accounted for almost a quarter of Rochester, New York's city payroll at its peak. Toyota employs a large proportion of its workforce in its headquarters city, Toyota. Even in a large city like Seattle, software engineers are at the mercy of Microsoft and Amazon. With these giants, a shock to a single firm is a shock to the entire market. If Microsoft has a bad year and lays off a large proportion of its software engineers, all software engineers in Seattle are hurt. How does this exposure to firm-specific shocks shape where workers and firms are located? And can the government use place-based policies to make everyone better off?

In this paper, we study these questions theoretically, empirically, and quantitatively. The key mechanism driving our results is a form of labor market pooling. If there is one firm in a labor market, and that firm has a negative shock, workers have nowhere to go. They are stuck at that unproductive firm. In contrast, if there are many firms in a market and one small firm becomes less productive, workers can move to another firm that is doing better and use their skills more productively. Thus, markets with many firms provide a "constant market for skill" as Marshall (1920) said and Krugman (1992) formalized. This labor market pooling mechanism implies that markets with many firms use labor more effectively than markets with a few dominant firms. Thus, when firms are granular, larger markets are more productive.[1]

Our primary contribution is a new, tractable model of local labor markets with endogenous entry of granular firms that is rich enough to allow for normative, empirical, and quantitative analysis. The model remains tractable because we introduce a continuum of sectors in each location and allow each sector to have a finite number of (ex-ante heterogeneous) firms. We then assume that firm entry is not targeted towards a particular sector, but towards an entire location. This ensures that workers stuck in a sector are exposed to individual firm shocks in their sector. However, firm entry does not devolve into intractable inequality constraints or multiple equilibria.[2]

We start by presenting a model of a single location with an exogenous population. Firms freely enter, workers choose what sector they would like to work in, and then each firm is subject to a productivity shock and hires labor in its sectoral labor market. We show that, in this setting, average wages are increasing in population because firms use labor more effectively in large labor markets. In particular, firms in large markets expand their employment more in response to productivity shocks, and so they use more workers while they are productive. Therefore, average labor productivity is higher. To see why, consider a local sectoral labor market with only one firm. Because workers are stuck in the short run, the firm does not change its employment in response to changes in productivity. Instead, it only drives up the market wage when it demands more

[1] In this paper, when we say firms are granular, we simply mean that they are not measure zero. Each firm is a grain with its own mass.

[2] See Bresnahan and Reiss (1991) for a discussion of the issues that arise in a model of firm entry with a finite number of firms.

labor. In contrast, if a firm hires a small share of the local sectoral labor market because the market is very large, when it increases its demand for labor, the market wage only increases a little, and the firm expands. We also show that the marginal benefits of increasing the size of the region disappear as the market becomes very large. Once a market is sufficiently large, no single firm can drive up the market wage. Therefore, increasing the size will not further improve productivity.

We go beyond this intuition and show that a statistic for how effectively a single firm uses labor is the covariance of that firm's log productivity and its log employment. That is to say, a firm uses labor more productively if it increases its employment when it has high productivity and decreases its employment when it has low productivity. The productivity of the entire market is just the average (employment-weighted) value of that covariance across firms. Thus, we can confirm our intuition of how the mechanism operates and quantify the contribution of granularity to the wage premium of large cities in a transparent, theory-consistent way.

Our last theoretical result considers the implications for policy. Our mechanism implies that too few firms enter in equilibrium. That is because, when firms are granular, they know that their entry affects wages. Not only will their entry increase average wages in a region, but it will increase wages precisely when the firm would like to hire more workers because its own attempt to increase its employment drives up the wages. That induces a correlation between wages and idiosyncratic firm shocks, which depresses firm profits since profit functions are convex in wages and productivity. Therefore, firms will "under-supply" their entry. A social planner can achieve a Pareto improvement using place-based subsidies on firm entry, especially in small locations where granularity matters.

We then enrich the model to test some of the predictions and to quantify the importance of our theoretical mechanism. We introduce imperfect mobility of labor across both firms and sectors, and we allow firms to internalize their market power, competing against each other à la Cournot. We embed this model into a standard model of economic geography in which workers are mobile across locations. Our main theoretical results hold in this richer setting when firms are competitive, but they need to be adjusted when firms are imperfectly competitive. As we show, imperfect competition strengthens the agglomeration benefits of granularity. However, the optimal policy is different. The social planner still provides larger subsidies to smaller locations, but that policy might be targeted towards workers in terms of wage subsidies rather than firms in terms of entry subsidies.

Our quantitative and empirical analysis focuses on Japan, where we have a rich panel of all manufacturing establishments with at least 4 employees every year.[3] These data include employment, payroll, and shipments by 6-digit product category. [4] We define a sector in the model as a 3-digit JSIC industry and a location as a commuting zone. We then estimate the key parameters of

[3]For the analyses using establishment-level data, we restrict our samples to establishments with at least 10 employees.

[4]We focus on Japan because the availability of this annual panel of near-universe establishments is necessary for our analyses. Some readers may think Japan is not the best setting for our mechanism, and it likely matters more for other countries with more fluid labor markets, like the United States. Perhaps surprisingly, that is not the case, and using the US parameters from the literature does not change the magnitude of our mechanism.

the model using these data. In an important way, Japan is not an ideal setting for our mechanism *quantitatively* because the labor market is relatively illiquid. We use Japan because the data allow us to test the key predictions and estimate our model, but we view our results as a lower bound on the importance of the mechanism in developed countries.

Before we use the estimated model to quantify our mechanism, we look to validate the model, both qualitatively and quantitatively, by considering some reduced-form evidence of our mechanism. We start by providing evidence that granularity matters. We show that the variance of log payroll in a local sectoral labor market is decreasing in the number of firms in that sector, suggesting, consistent with Gabaix (2011), that individual firms are subject to idiosyncratic shocks. And those shocks average out in larger markets.

We then provide evidence consistent with our mechanism, as suggested by the covariance statistic. We begin by showing that the variance of log employment at a single firm in a large local sectoral labor market is larger on average than that of a similarly situated firm in a small local sectoral labor market. That suggests that if those firms are subject to similar shocks, the firm in the larger market expands more in response to good shocks and shrinks more in response to bad ones. We also provide evidence that this is due to our mechanism rather than another mechanism, for instance, increasing returns to scale in the matching function. To provide more direct evidence, we construct revenue productivity shocks to each firm using that firm's exposure to national changes in demand for the products it produces. Consistent with our mechanism, both qualitatively and quantitatively, firms that already hire a large portion of the local sectoral labor market expand their employment less in response to these shocks.

Finally, we turn to demonstrating the quantitative importance of the mechanism. If firms are perfectly competitive, we find that the implied elasticity of wages to population gets as high as 0.0074 in the smallest locations, and is a much smaller 0.0009 in Tokyo. In total, the mechanism implies that Tokyo is 1.5% more productive than the smallest city in Japan. If firms are Cournot competitive, the implied elasticity of wages to population reaches 0.0068 in the smallest locations. Combes et al. (2011) find that most causal estimates of the urban wage premium are an elasticity between 0.02 and 0.05 when pooling across locations of all sizes. Therefore, granularity could explain as much as 37% ($= 0.0074/0.02$) of the urban wage premium.

We then quantify the implications for optimal policy. We define the marginal product of labor and firms at the commuting zone level as the marginal contribution of another worker or firm to the output of the entire commuting zone. We then compare those values to the wages and profits in each location to see if workers and owners are properly compensated for their contributions. The results depend on assumptions about firm conduct. If firms are perfectly competitive, workers are paid their marginal product, and firm profits are only 96.7% of their marginal product in small locations. In large locations, firms capture 99.6% of their contribution to commuting zone production. Putting in place the optimal firm entry subsidies would increase the number of firms in the smallest locations by 3.4% and in Tokyo by 0.5%. This reallocation of firms leads a small number of workers to move, increasing the population in small areas by about 0.5%.

If firms are Cournot competitive in labor markets, workers in small locations see an average wage markdown of 0.52%. This implies that there is over-entry of firms in the largest locations, while too few firms enter the smallest locations. The optimal policy also features a wage subsidy, especially in the smallest locations, to undo the wage markdown. On net, these policies end up increasing the population in the smallest cities by about 0.6%.

The rest of the paper is organized as follows. We give a literature review below. In Section 2, we present the baseline model of a single location and show our theoretical results. We enrich the model in Section 3 and estimate it in Section 4. We validate the model and our mechanism in Section 5 before presenting the quantitative results in Section 6. Section 7 concludes and suggests ways the model could be enriched to capture important real-world features of granular markets.

Related Literature. The literature on the spatial agglomeration of economic activity is rich. In an early contribution, Marshall (1920) proposes three reasons why firms might locate around other firms: labor market pooling, access to intermediates, and the sharing of ideas. Subsequent theory papers have formalized these ideas and offered other potential mechanisms (Miyauchi, 2024; Davis and Dingel, 2019). Duranton and Puga (2004) provide a new way to classify these mechanisms in their review. Many empirical studies have shown that there are benefits from agglomeration (Andersson et al., 2014; Kline and Moretti, 2014; Greenstone et al., 2010) and have also analyzed the co-agglomeration patterns of sectors to infer the relative importance of different theoretical mechanisms (Ellison and Glaeser, 1997; Ellison et al., 2010). Rosenthal and Strange (2004) review the evidence.

Our paper focuses on a particular mechanism that falls under the broad umbrella of labor market pooling. There are many microfoundations with different mechanisms of how labor market pooling can lead to agglomeration benefits (Andersson et al., 2007; Papageorgiou, 2022). Our model builds on the basic theoretical framework of Krugman (1992), which considers a setting with a finite number of ex-ante identical firms where labor is perfectly mobile within a labor market but not across. Overman and Puga (2010) extend Krugman's model to include multiple sectors and test the predictions about where those sectors should be located. Other papers test these predictions in different settings (de Almeida and de Moraes Rocha, 2018; Nakajima and Okazaki, 2012). We provide a new model with ex-ante heterogeneous firms and a continuum of sectors that yields new theoretical results clarifying how granular firms generate agglomeration. We also derive the normative implications. Beyond the theoretical contribution, our model can be applied to the data to provide direct evidence of the mechanism and quantify its importance, which has been previously impossible with the stylized models that exist in the literature.

More recent work looks for direct evidence of the labor market pooling mechanism. Moretti and Yi (2024) show that workers who are laid off in large labor markets have an easier time finding work compared to workers in small markets. This is consistent with the theory we lay out, though our evidence focuses on the firm response rather than the worker side. Conte et al. (2024) show that when firms in large markets can more easily expand in response to productivity shocks, more

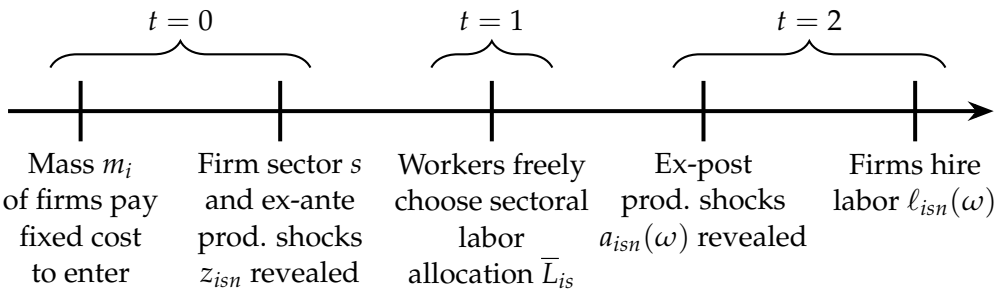


Figure 1: Timing of model

volatile firms will sort into larger markets. Our baseline model abstracts from firm sorting but demonstrates how granularity could explain why firms in larger markets can expand more in response to productivity shocks.

We build on a large literature recently inspired by Gabaix (2011) that looks to quantify what the granular nature of firms means for economic activity and optimal policy. Gabaix (2011) shows that shocks to individual firms could explain nationwide fluctuations. Bernard et al. (2018) give a framework for thinking about how a few important firms could shape the nature of international trade. Gaubert and Itskhoki (2021) discuss what granularity means for the observed comparative advantage of countries, and Gaubert et al. (2021) study what that implies for optimal policy. Schoefer and Ziv (2024) study how much of the observed differences across locations can be explained by granular shocks. We contribute to this literature by considering granularity's implications for the geography of economic activity and optimal place-based policy.

2 How Does Granularity Drive Agglomeration?

The goal of this section is to demonstrate how granularity leads to higher wages in larger markets through the labor market pooling mechanism discussed by Marshall (1920). To isolate the effects of granularity most transparently, we focus on the sectoral labor markets of a single location and consider a simplified environment that is neoclassical, conditional on firm entry.

We will enrich the model by endogenizing where people live, allowing for imperfect mobility across firms and sectors, and introducing imperfect competition in Section 3. Throughout the paper, we assume that workers and firms are homogeneous before making any training or entry decisions. Allowing for specialized skills used across different sectors and directed firm entry across regions and sectors is outside the scope of the current study. We extend the baseline model to include firm sorting and capital in Appendices C.3 and C.4. The basic results we show in the main text do not change much, qualitatively or quantitatively.

2.1 Environment

There is a region i with a mass ℓ_i of workers and a continuum of sectors $s \in [0,1]$. The sectors produce perfectly substitutable goods but hire in distinct sectoral labor markets.

Timing. There are three periods $t \in \{0,1,2\}$. In period 0, a mass m_i of firms pay a fixed cost of the traded final good in order to enter. Each firm is randomly assigned a sector s and then gets an ex-ante productivity draw z from some known distribution. Thus, each sector ends up with a finite number of firms N_{is} that differ in their productivity even though a non-integer mass of firms enter location i. This captures the long-run differences in firm size and will determine how exposed different sectoral markets are to short-run idiosyncratic firm shocks in period 2.

After observing those initial productivity draws, a representative worker freely allocates her labor $\overline{L}_{is}$ across sectors s in period 1. This captures the long-run decision of a worker free to direct her search or make training choices toward certain sectors in her region.

Then, in period 2, the state of the world $\omega \in \Omega$ is revealed. This determines the short-run productivity shocks of each firm. The worker cannot move labor across sectors at this time.[5] Instead, she supplies her sectoral labor inelastically, and firms hire labor in the sectoral labor market. Firms then produce and sell their goods. The fact that firms are better able to respond to these period 2 productivity shocks in larger markets will imply the agglomeration benefits. The model timing is summarized in Figure 1.

Workers. In location i, risk-neutral representative agents get utility from consuming a freely traded final good c_i and are endowed with one unit of labor that they supply to the market inelastically. In period 1, a worker freely allocates her units of labor across sectors $s \in [0,1]$, taking as given the number of firms and each firm's ex-ante productivity z_{isn}. In particular, she chooses her vector of labor supply $\boldsymbol{L_i} \equiv \{L_{is}\}_s$ in the set of feasible labor allocations $\mathcal{L}$, i.e.

$$\boldsymbol{L_i} \in \mathcal{L} \equiv \left\{ \boldsymbol{L_i'} | \int_0^1 L_{is}' ds \leq 1 \right\}.$$

In period 2, the state of the world ω is revealed. The worker is unable to adjust labor at this point, and so inelastically supplies L_{is} labor to sector s.

Firms. There is a continuum of potential firm entrants. To enter, a firm must pay a fixed cost $\psi_i > 0$ in terms of the freely traded final good in period 0.[6] Those firms are then randomly assigned a sector. We denote by $\mathcal{N}_{is}$ the set of firms operating in region i sector s and $N_{is} \equiv |\mathcal{N}_{is}|$ the (finite) number of firms. We assume that firms enter according to the "ball-and-urn model" so that N_{is} is distributed Poisson with mean m_i. That is, the probability mass function for the number

[5] This assumption is made for clarity; none of the key results depend on it. We allow for imperfect movement across both firms and sectors in period 2 in Section 3.

[6] In Appendix C.2, we consider an alternate model where the fixed cost is paid in terms of local labor. The results do not substantively change.

of firms in a sector is $m_i^N e^{-m_i}/N!$. Firm n in sector s then gets an ex-ante productivity draw z_{isn} from a distribution $F_{iz}(\cdot)$.

In period 2, each firm n gets an ex-post idiosyncratic productivity shock, $\tilde{a}_{isn}(\omega)$, a sector-wide productivity shock, $\tilde{A}_{is}(\omega)$, and produces a final good, $y_{isn}(\omega)$, according to,

$$y_{isn}(\omega) = z_{isn} a_{isn}(\omega) \ell_{isn}(\omega)^{1-\eta},$$

where $a_{isn}(\omega) \equiv \tilde{a}_{isn}(\omega)\tilde{A}_{is}(\omega)$ is the total productivity shock to firm n, $\ell_{isn}(\omega)$ is the total amount of labor firm n hires, and $\eta \in (0,1)$ is the degree of decreasing returns to scale. We formally state the assumptions on the productivity distributions below.

Assumption 1. The productivity distributions satisfy the following properties:

- Absent ex-post shocks, the expected firm size is finite, i.e. $\mathbb{E}[z_{isn}^{1/\eta}]$ exists and is finite;
- $\log \tilde{a}_{isn}(\omega)$ are iid with mean zero and a finite second moment σ_N^2; and
- $\log \tilde{A}_{is}(\omega)$ are iid with mean zero and a finite second moment σ_S^2.

Market Clearing. Total expected production in the location is

$$Y_i = \mathbb{E}\left[\int_0^1 \sum_{n \in \mathcal{N}_{is}} z_{isn} a_{isn}(\omega) \ell_{isn}(\omega)^{1-\eta} ds\right],$$

where the expectation is taken with respect to ω, the number of firms in each sector, and the ex-ante productivity draws. We assume that the law of large numbers applies across this continuum of sectors so that realized production is always Y_i. Therefore, goods market clearing requires that total consumption plus the amount of the final good used for investment must equal expected production,

$$c_i \ell_i + \psi_i m_i = Y_i. \tag{1}$$

In the labor market, labor demanded equals the individual labor supplied by each worker multiplied by the number of workers,

$$\sum_{n \in \mathcal{N}_{is}} \ell_{isn}(\omega) = L_{is} \ell_i, \quad \forall s, \omega. \tag{2}$$

2.2 Market Structure and Equilibrium

Labor Supply. Workers choose their labor allocation across sectors in period 1 to maximize their expected utility, taking wages as given. We normalize the price of the final good to 1, so workers solve the problem.

$$\boldsymbol{L_i} \in \underset{\boldsymbol{L_i'} \in \mathcal{L}}{\operatorname{argmax}} \quad \mathbb{E}\left[\int_0^1 w_{is}(\omega) L_{is}' ds\right], \tag{3}$$

where $w_{is}(\omega)$ is the equilibrium wage for sector s in state of the world ω. We will denote the maximum of (3) by w_i.[7]

Labor Demand. We normalize productivity so that the price of each good is 1. Then each active firm maximizes profits, taking wages and prices as given,

$$\ell_{isn}(\omega) \in \underset{\ell'}{\text{argmax}} \quad z_{isn} a_{isn}(\omega)(\ell')^{1-\eta} - w_{is}(\omega)\ell'. \tag{4}$$

We will denote the maximum of (4) by $\pi_{isn}(\omega)$. We assume that firms are price takers, even though they are large relative to their market, for simplicity. In Appendix A.4, we show that nothing changes if firms commit to wage schedules in period 1 and compete à la Bertrand.

Entry. We assume that firms enter up to the point that expected profits are equal to the fixed cost of entering. After those entry decisions are made, all firms are randomly assigned to their sector and get their productivity draws. We can write this,

$$\psi_i = \frac{\mathbb{E}\left[\int_0^1 \sum_{n \in \mathcal{N}_{is}} \pi_{isn}(\omega) ds\right]}{m_i}. \tag{5}$$

The numerator is the total amount of profits earned by firms in location i. The denominator is the total measure of firms that enter, so the ratio is a firm's expected profit before any firm realizes its sector or productivity shocks.[8]

Definition 1. A *local equilibrium* consists of wages $w_{is}(\omega)$, labor supply decisions L_{is}, entry decisions m_i, and labor demand decisions $\ell_{isn}(\omega)$, such that

- Workers make labor supply decisions to maximize utility, taking wages as given, (3);
- Conditional on entry, firms maximize profits taking prices and wages as given, (4);
- Firms enter up to the point that expected profits are equal to the fixed cost of entering, (5); and
- Goods and labor markets clear, (1) and (2).

2.3 Labor Market Pooling and Agglomeration

We now demonstrate how granularity implies that average wages increase with the number of workers. We will proceed in two steps. First, in this subsection, we will show that average wages increase with the number of workers if firms adjust their employment more in response to

[7] In equilibrium, this will imply that average wages are equalized across sectors.

[8] An alternate entry decision would allow the marginal firm to observe the distribution of firms across sectors and their ex-ante productivity shocks before deciding whether or not to enter. We show that the free entry condition implied by that alternate entry game is also (5) if firms internalize that, were they to enter a sector with N firms, the sector would have $N+1$ firms in Appendix C.1.

productivity shocks in larger markets. Throughout this paper, we will refer to this mechanism as labor market pooling, as it implies firms adjust their employment more in response to the same shock in a larger market.

Second, in Section 2.4, we will demonstrate how the existence of granular firms implies that firms expand their employment more in response to shocks in larger markets. Combining this with Section 2.3, we conclude that granularity implies agglomeration benefits. Finally, in Section 2.5, we present the implications for optimal policy.

For tractability, we will derive all of our results using a log second-order approximation to expected production around the point with no ex-post shocks. We will use $\overline{x}$ to denote the value of a variable x in the absence of any ex-post shocks.

We start by introducing the *Regional Production Function* as it will be useful to organize the discussion. The regional production function gives the maximum possible production for location i, taking as given the mass of firms m and the number of workers ℓ:

$$Y_i(\ell, m) \equiv \max_{\ell'_{sn}(\omega), L_s} \left\{ \mathbb{E}\left[\int_0^1 \sum_{n \in \mathcal{N}_s} z_{sn} a_{sn} \ell'_{sn}(\omega)^{1-\eta} ds \middle| m \right] \middle| \boldsymbol{L}' \in \mathcal{L}, \sum_{n \in \mathcal{N}_s} \ell'_{sn}(\omega) = L'_s \ell \right\}.$$

Since labor markets are perfectly competitive, this is not only the maximum level of production, but it also corresponds to the equilibrium production at the equilibrium levels of m and ℓ. The only difference between the regional production function and equilibrium production is that the number of firms is endogenously determined in equilibrium, while they are taken as given in the regional production function.

In the following Lemma, we characterize $Y_i(\ell, m)$ up to log second order.

Lemma 1. *The regional production function is* $Y_i(\ell, m) = z_i m^\eta \ell^{1-\eta} \Phi(m)$, *where* $z_i \equiv \mathbb{E}[z_{isn}^{1/\eta}]^\eta$ *and* $\Phi(m)$ *is given by,*

$$\Phi(m) \equiv \mathbb{E}[a_{sn}(\omega)] + \frac{1-\eta}{2} \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n \in \mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathrm{Cov}_\omega \left(\log a_{sn}(\omega), \log \ell_{sn}(\omega) \right) ds, \tag{6}$$

where the covariance is for a single firm across different states of the world.

The formal proof of Lemma 1 is in Appendix B, but we will provide a quick sketch here. We start by solving the maximization problem with no ex-post shocks. We then do a second-order approximation to the maximand, which implies $Y_i(\ell, m) = z_i \ell^{1-\eta} m^\eta \tilde{\Phi}(m)$ where

$$\begin{aligned} \tilde{\Phi}(m) \quad \equiv \quad & \mathbb{E}[a_{sn}(\omega)] + (1-\eta) \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n \in \mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathrm{Cov}\left(\log a_{sn}(\omega), \log \ell'_{sn}(\omega) \right) ds \\ & - \eta \frac{1-\eta}{2} \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n \in \mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathrm{Var}\left(\log \ell'_{sn}(\omega) \right) ds. \end{aligned} \tag{7}$$

That is, relative to keeping labor at its no-shock level in every state of the world, there are gains from increasing labor at firms that have good productivity shocks and decreasing labor at

firms with bad productivity shocks. Those gains are tempered by the fact that there are decreasing returns to scale, so that productivity decreases if there is a very high variance of $\log \ell'_{sn}(\omega)$. When labor is efficiently allocated, either because the market is competitive or there is a planner,

$$\eta \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n \in \mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \operatorname{Var}\left(\log \ell'_{sn}(\omega)\right) ds = \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n \in \mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \operatorname{Cov}\left(\log a_{sn}(\omega), \log \ell'_{sn}(\omega)\right) ds, \tag{8}$$

proving the result. This leads to our first substantive result.

Corollary 1. *The regional production function features increasing returns to scale, i.e.,* $\frac{dY_i(\alpha\ell,\alpha m)}{d\alpha} > Y_i$*, if and only if the employment-weighted covariance between productivity and employment is increasing in the number of firms, i.e.,* $\frac{\partial \Phi(m)}{\partial m} > 0$.

This result follows immediately from Lemma 1 since

$$\frac{dY_i(\alpha\ell, \alpha m)}{d\alpha} = \frac{d}{d\alpha}\left[\alpha z_i m^{\eta} \ell^{1-\eta} \Phi(\alpha m)\right] = Y_i(m, \ell)\left(1 + \frac{\partial \log \Phi(m)}{\partial \log m}\right).$$

Corollary 1 provides a clear interpretation of how labor market pooling implies that the regional production function features increasing returns to scale. Consider a firm n that never adjusts its workforce in response to idiosyncratic productivity shocks so that the covariance is 0. That firm would hold onto a large number of workers when it has bad productivity shocks and not expand to take advantage of good productivity shocks. Therefore, its average labor productivity would depend solely on its average productivity shock. By contrast, if the firm were to expand after a good productivity shock and shrink after a bad shock, its average labor productivity would increase because it hires more workers when it is more productive. Thus, that firm could produce more goods on average while hiring the same average number of workers simply by increasing its covariance.

Corollary 1 then says that there are increasing returns to scale if increasing the size of the market improves how much labor reallocates across firms in response to productivity shocks, properly weighted by the importance of each firm. Thus, if firms are better able to expand their employment in response to good productivity shocks in a larger market, then larger markets will be more productive.

Relation to Misallocation. This mechanism is closely related to the misallocation literature (Hsieh and Klenow, 2009). To see this in the math, note that for any firm on its labor demand curve, $w_s(\omega) = (1-\eta) z_{sn} a_{sn}(\omega) \ell_{sn}(\omega)^{-\eta}$. Therefore, looking at the variance of log wages,

$$\int_0^1 \frac{\overline{\ell}_s}{\overline{\ell}} \operatorname{Var}(\log w_s(\omega)) ds = \operatorname{Var}(\log a_{sn}(\omega)) - \eta \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n \in \mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \operatorname{Cov}(\log a_{sn}(\omega), \log \ell_{sn}(\omega)) ds,$$

using equation (8). That is, the employment-weighted variance of log wages is decreasing in the employment-weighted covariance of productivity and labor. This immediately implies the next

corollary.

Corollary 2. *The regional production function features increasing returns to scale* $\frac{dY_i(\alpha\ell,\alpha m)}{d\alpha} > Y_i$ *if and only if the employment-weighted variance of log wages is decreasing in the number of firms.*

Corollary 2 says that larger markets are more productive than smaller markets if the average variance of log wages is lower in large markets. Loosely speaking, this is because there is less "misallocation" of labor. More precisely, the variance of log wages in sector s is a measure of how much more productive labor in sector s is in some states of the world than in others. Therefore, it can be thought of as a measure of how unproductively labor is used on average.

Importantly, unlike misallocation, this variance is not a sign of inefficiency. Markets are perfectly competitive and are therefore efficient. Workers would love to reallocate some of their labor from states of the world in which the sectoral wage is low to states of the world where wages are high if they could. But that is not possible because the labor is stuck in each sector across all states of the world.

Agglomeration Benefits. We next turn to show how the increasing returns to scale of the regional production function $Y_i(\ell, m)$ imply that there are benefits to agglomeration. To do that, we need to relate the equilibrium number of firms to $Y_i(\ell, m)$. This is straightforward as wages are set competitively, so workers are paid their marginal product. That is,

$$w_i = \frac{\partial Y_i}{\partial \ell} = \frac{(1-\eta)Y_i}{\ell_i}. \tag{9}$$

Therefore, total profits are simply $Y_i - w_i\ell_i = \eta Y_i$, so that free entry (5) can be rewritten,

$$\psi_i = \frac{\eta Y_i}{m_i}. \tag{10}$$

Throughout the paper, we will focus on equilibria where the degrees of increasing returns to scale are small enough that profits are decreasing in the number of firms.

Combining equations (9) and (10) with Corollary 1 then implies the next proposition.

Proposition 1. *Average wages are increasing in the number of workers, i.e.* $\frac{d\log w_i}{d\log \ell_i} > 0$*, if and only if the employment weighted covariance between log firm productivity and log firm employment is increasing in* m*, i.e.* $\frac{\partial \log \Phi(m)}{\partial \log m}\Big|_{m=m_i} > 0$.

The proof proceeds in two steps. Equation (10) implies that the mass of firms increases more than proportionally with the increase in labor, $\frac{d\log m_i}{d\log \ell_i} = \frac{1-\eta}{1-\eta-\frac{\partial \log \Phi(m)}{\partial \log m}} > 1$, as average productivity

increases with more entry.[9] Then equation (9) implies average wages increase,

$$\frac{d \log w_i}{d \log \ell_i} = \frac{\frac{\partial \log \Phi(m)}{\partial \log m}}{1 - \eta - \frac{\partial \log \Phi(m)}{\partial \log m}} > 0.$$

Proposition 1 summarizes the basic argument of labor market pooling. If firms expand more in response to good productivity shocks in larger markets than in smaller markets, then average wages will increase with the size of the market.

2.4 Granularity and Labor Market Pooling

The previous subsection demonstrated how labor market pooling mechanisms can make larger markets more productive than smaller markets. In this subsection, we complete the argument that granularity implies agglomeration benefits by demonstrating how granularity implies those labor market pooling forces.

Proposition 2. *The average wage is increasing in the number of workers, i.e.,* $\frac{d \log w_i}{d \log \ell_i} > 0$*, if and only if idiosyncratic shocks have a positive variance,* $\sigma_N^2 > 0$*. Furthermore, the agglomeration benefits converge to zero as the size of the market goes to infinity, i.e.* $\frac{d \log w_i}{d \log \ell_i} \to 0$ *as* $\ell_i \to \infty$*.*

We start by giving a basic intuition for why there are increasing returns to scale using Proposition 1. Consider how firm n in sector s responds to an idiosyncratic, ex-post productivity shock $\Delta \log a_{isn}(\omega)$. To first order, for a firm on its labor demand curve,

$$\Delta \log \ell_{isn}(\omega) = \frac{1}{\eta} \left(\Delta \log a_{isn}(\omega) - \Delta \log w_{is}(\omega) \right) = \frac{1}{\eta} \left(1 - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \right) \Delta \log a_{isn}(\omega),$$

where the last equality follows from the fact that the pass-through of a firm's productivity shock to sectoral wages is given by the share of the labor force that the firm hires $\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}$.

In a small region, where the mass of firms m_i is small, chances are that there are very few other firms in sector s. Therefore, firm n hires a large share of the labor force in sector s, i.e. $\overline{\ell}_{isn}/\overline{\ell}_{is}$ is large. In that case, the firm's employment does not respond much after a productivity shock because it already hires a large proportion of the sectoral labor force, and it cannot attract workers from other sectors. Instead, it drives up the wages in the sector. Therefore, the firm does not effectively scale up in response to a productivity shock and ends up using labor unproductively. In a market with a large mass of firms m_i, firm n's share of the sector s labor force is smaller. Therefore, firm n's labor responds more in response to productivity shocks because it can hire workers away from other firms in the sector, using that labor more productively.

[9] This follows from our assumption that the fixed cost is paid in terms of the freely traded final good. If the fixed cost is paid in local labor, the number of firms increases proportionally with labor. In Appendix C.3 we allow for firm sorting and find that the number of firms increases less than proportionally with labor, in line with what we see in the data.

The formal proof for Proposition 2 is in Appendix B, but we give a sketch of the proof here. Allowing for productivity shocks to every firm in sector s implies that, to first order, labor at firm n in sector s is given by

$$\Delta \log \ell_{isn}(\omega) = \frac{1}{\eta}\left(\Delta \log a_{isn}(\omega) - \sum_{n' \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}} \Delta \log a_{isn'}(\omega)\right).$$

Then, some straightforward algebra reveals that the employment weighted average covariance is given by

$$\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \operatorname{Cov}(\log a_{isn}(\omega), \log \ell_{isn}(\omega)) ds = \frac{1}{\eta}\left(1 - \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} HHI_{is} ds\right) \sigma_N^2,$$

where $HHI_{is} \equiv \sum_{n \in \mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2$. This covariance depends on the number of firms only through the HHI_{is}. Thus, the degree of increasing returns to scale is

$$\frac{\partial \log \Phi(m)}{\partial \log m} = -\frac{1}{2}\frac{1-\eta}{\eta}\frac{\frac{\partial}{\partial \log m}\left[\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} HHI_{is} ds\right]}{\Phi(m)} \sigma_N^2. \tag{11}$$

If the average *HHI* across sectors decreases as the number of firms m increases, Proposition 1 is satisfied, and there are agglomeration benefits. We prove that the *HHI* of a sector s is decreasing in the number of firms N_{is}, so the last thing we need to check is that firm entry across sectors is not too strange so that average HHI across sectors in location i is decreasing in the mass of firm entrants, m_i. For example, suppose that there were only two sectors with firms in them: one with two firms and another with one firm. If firm entry goes into unoccupied sectors, the average *HHI* would actually increase as the average number of firms in occupied sectors decreases. This, and other pathological cases, are ruled out with Poisson entry.

The speed with which the average *HHI* decreases depends on the distribution of entrants across sectors and the ex-ante productivity distribution $F_{is}(z)$. Entry especially matters for low m. As m becomes larger, the ex-ante productivity distribution matters more. As discussed in Gabaix (2011), if $F_{is}(z)$ has thin tails, HHI_{is} decreases approximately at the rate of N_{is}^{-1}. Then the granular agglomeration forces are strong when there is a small number of firms and the *HHI* falls quickly with new entrants. However, the *HHI* quickly approaches zero, at which point the average *HHI* cannot fall further. For example, if a sector has one firm and adds another ex-ante identical firm, the HHI drops from 1 to 0.5. If a sector already has 100 identical firms, the *HHI* is 0.01, doubling the number of firms only decreases it to 0.005, not increasing productivity much. Intuitively, that is because if there are already 100 firms in a sector, it is easy for any firm to expand by attracting workers from the other 99 firms. Adding more firms does not have much effect.

If $F_{is}(z)$ has thick tails, HHI_{is} decreases at a rate of $N_{is}^{-\xi}$ where $\xi \in (0,1)$.[10] In that case, the weighted *HHI* does not fall as quickly, and large firms continue to be constrained in their ability to

[10]Formally, if $F_{is}(z)$ is a Pareto distribution, i.e. $F_{is}(z) = 1 - az^{-\lambda}$, it is thick-tailed if $1 < \lambda\eta < 2$. Then $\xi =$

respond to productivity shocks in larger markets. Thus, the externality is not as strong for small markets, but matters for medium-sized and even large cities. Nonetheless, the agglomeration force becomes less important as the number of firms increases, and the average *HHI* approaches zero.

2.5 Optimal Policy

In this subsection, we consider what this force for agglomeration means for policy. Conditional on firm entry, the model is competitive, so the only possible source of inefficiency is firm entry. Rewriting the goods market clearing condition (1), the first-best level of entry m_i^{FB} maximizes the total production net of firm entry costs,

$$m_i^{FB} \in \underset{m'}{\text{argmax}} \quad Y_i(\ell_i, m') - \psi_i m'. \tag{12}$$

Taking the first order condition, we find that the first-best entry must satisfy

$$\psi_i = \left(1 + \frac{1}{\eta} \frac{\partial \log \Phi\left(m_i^{FB}\right)}{\partial \log m}\right) \frac{\eta Y_i}{m_i^{FB}}.$$

In contrast, equilibrium entry satisfies equation (10). Therefore, to implement the first best entry, the planner needs to enact a firm entry subsidy proportional to profits given by $\tau_i = \frac{1}{\eta} \frac{\partial \log \Phi(m)}{\partial \log m}$. And as $\frac{\partial \log \Phi(m)}{\partial \log m} \to 0$ as $m \to \infty$, the size of the optimal subsidy will also converge to zero in large markets. This implies our normative result.

Proposition 3. *If idiosyncratic shocks have a positive variance, $\sigma_N^2 > 0$, the optimal policy features a subsidy on firm entry proportional to firm profits given by $\tau_i = \frac{1}{\eta} \frac{\partial \log \Phi(m)}{\partial \log m}\big|_{m=m_i} > 0$. Furthermore, the optimal subsidy converges to zero as the size of the market goes to infinity, i.e. $\tau_i \to 0$ as $\ell_i \to \infty$.*

One might wonder why there is any need for policy, given that wages are set competitively. The First Welfare Theorem fails in our model because firm entry is not Walrasian. In a competitive equilibrium, potential entrants take prices and wages as given when deciding whether or not they can make a profit by entering. In our model, by contrast, firms ask if current operating firms make a profit when deciding whether or not to enter. This condition is equivalent to one where a potential entrant internalizes that if it were to enter a sector with N firms, the sector would have $N + 1$ firms.[11]

Since firms are large relative to their sector, they know their entry will affect the wage distribution. A potential entrant understands that wages would be high precisely when it would like to hire more workers, since its own idiosyncratic shocks move market labor demand, and thereby, sectoral wages. Anticipating this endogenous wage response, a potential entrant is less likely to

$2\left(1 - \frac{1}{\lambda\eta}\right)$.

[11] We show the equivalence in Appendix C.1.

enter than a similarly situated Walrasian firm that takes the equilibrium, uncorrelated distribution of wages as given, since firm profit functions are convex. This under-entry force is strongest in small markets, where the marginal entrant has a large impact on the wage distribution.[12]

3 A Quantitative, Granular Model of Economic Geography

In Section 2, we presented a stylized model of sectoral labor markets within a single region where workers were freely mobile within a sector and unable to move across sectors. In this section, we model a country made up of I regions indexed by $i \in \mathcal{I} \equiv \{1, \ldots, I\}$. We do this by introducing a migration decision at time $t = -1$, allowing workers to choose where to live. We then assume that workers are stuck in their location for the remaining periods.

We extend the model of the labor market in Section 2 so that in period 1, workers allocate their labor across sectors and firms in their location. Then, in period 2, workers can move their labor across both firms and sectors, subject to movement frictions. We further allow firms to internalize their labor market power so that they compete with other firms in their sector according to Cournot competition.[13] We show that the main results remain unchanged in this extended model with perfect competition, and the results only need to be adjusted slightly with imperfect competition. A more detailed description of the model, a characterization of the equilibrium, and proofs of the results are in Appendix A.

3.1 Extending the Baseline Model

Migration Across Regions. There is a mass ℓ of workers in the country. The fundamental utility of living in location i is $u_i = \overline{u}_i w_i$, where $\overline{u}_i$ are the local amenities. Each worker has an idiosyncratic preference for each location ε_i so that the utility the worker gets from living in location i is $u_i \varepsilon_i$.

We assume that ε_i are distributed Fréchet with shape parameter $\theta > 0$. Therefore, when people are free to live where they would like, the number of people who live in location i is given by $\ell_i = (u_i/u)^\theta \ell$, where $u = \left(\sum_{i \in \mathcal{I}} (u_i)^\theta\right)^{\frac{1}{\theta}}$.

Imperfect Mobility across Firms and Sectors. In period 1, the representative worker freely allocates her units of labor across sectors $s \in [0,1]$ and firms $n \in \mathcal{N}_{is}$, taking as given each firm's ex-ante productivity z_{isn}. In particular, she chooses her vector of labor supply $\boldsymbol{L_i} \equiv \{L_{isn}\}_{s,n}$ in the set of feasible labor allocations $\mathcal{L}$,

$$\boldsymbol{L_i} \in \mathcal{L} \equiv \left\{ \boldsymbol{L_i'} | \int_0^1 \sum_{n \in \mathcal{N}_{is}} L'_{isn} ds \leq 1 \right\}.$$

[12] This result does not depend on our assumption that wages are set competitively. In Appendix A.4, we show that the under-entry result still holds if firms do not take wages as given and instead compete against each other à la Bertrand.

[13] In the appendix, we also present the model when firms compete in Bertrand.

In period 2, the state of the world ω is revealed. This determines the ex-post productivity shocks for all firms. The worker then reallocates labor across firms, choosing a vector of labor supply $\boldsymbol{L_i(\omega)} \equiv \{L_{isn}(\omega)\}_{s,n}$ in the set of feasible labor allocations $\mathcal{L}_\Omega(\boldsymbol{L_i})$ which depends on the worker's labor choices in period 1. The set is given by

$$\mathcal{L}_\Omega(\boldsymbol{L_i}) \equiv \left\{ \boldsymbol{L_i(\omega)} | 1 = \left(\int_0^1 L_{is}^{-\frac{1}{\nu}} L_{is}(\omega)^{\frac{1+\nu}{\nu}} ds \right)^{\frac{\nu}{1+\nu}}, \quad L_{is}(\omega) = \left(\sum_{n \in \mathcal{N}_{is}} \left(\frac{L_{isn}}{L_{is}} \right)^{-\frac{1}{\kappa}} L_{isn}(\omega)^{\frac{1+\kappa}{\kappa}} \right)^{\frac{\kappa}{1+\kappa}} \right\},$$

where $L_{is} = \sum_{n \in \mathcal{N}_{is}} L_{isn}$.

This set implies that $L_{is}(\omega) = L_{is}$ and $L_{isn}(\omega) = L_{isn}$ is feasible, but the more that the worker deviates from period 1 labor choices, the more labor is lost in transition. $\kappa > 0$ is the short-run elasticity of substitution across firms within a sector, and $\nu > 0$ is the short-run elasticity of substitution across sectors. The worker in location i chooses $\boldsymbol{L_i}$ and $\{\boldsymbol{L_i(\omega)}\}_\omega$ to maximize expected utility taking wages as given.

Ex-ante Productivity Distribution. The ex-ante productivity distribution is Pareto with shape parameter λ and location parameter ζ_i that can differ across locations. We assume that $\lambda\eta > 1$ so that the expected firm size is finite.

Labor Market Competition. For the rest of the paper, we will present results under alternate assumptions on firm behavior. Under perfect competition, firms continue to maximize profits, taking wages as given. However, now that workers are imperfectly substitutable across firms, each firm has its own wage $w_{isn}(\omega)$ that it takes as given.

Under imperfect competition, firms internalize that they can affect the wages they face. We assume that the firm can commit to a wage schedule in period 1, so that the firm internalizes how wage choices in state of the world ω will affect how much period 1 labor the worker supplies to the firm. This captures the decision-making of a firm that could lower wages significantly today but knows its workers would leave in the long run as a result. This implies that, in the absence of ex-post shocks, wages will correspond with competitive wages as workers are freely mobile in period 1. With ex-post shocks, firms will sometimes hire a larger portion of the market and have more market power. Therefore, the firm can vary its markdown across different states of the world to increase profits. We assume Cournot competition, so that each firm takes as given the labor hired by the other firms in its sector and the work opportunities in the other sectors.

3.2 Theoretical Results with Perfect Competition

When firms engage in perfect competition, the key results, Lemma 1, Proposition 1, Proposition 2, and Proposition 3 do not change at all as long as workers are more mobile across firms within a sector than across sectors, $\kappa > \nu$. The only qualitative change is that Corollary 2, relating the mechanism to misallocation, no longer holds. That is because misallocation only measures

how efficiently existing labor is used. With movement frictions, the worker also needs to consider how much labor is lost in moving labor across firms and sectors.

Quantitatively, numbers change as the covariance, determining the strength of agglomeration, is given by

$$\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \text{Cov}(\log a_{isn}(\omega), \log \ell_{isn}(\omega)) ds = \frac{1}{\eta + \frac{1}{\nu}} \sigma_S^2 + \frac{\eta + \frac{1}{\nu} - \left(\frac{1}{\nu} - \frac{1}{\kappa}\right) \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} HHI_{is} ds}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)} \sigma_N^2.$$

The size of the externality then also depends on how much harder it is for a firm to attract workers from other sectors compared to workers from other firms in its own sector, $\frac{1}{\nu} - \frac{1}{\kappa}$.

3.3 Theoretical Results with Imperfect Competition

In this subsection, we discuss how the theoretical results change when firms compete à la Cournot.

Most importantly, Lemma 1 does not hold with Cournot competition as the covariance expression relies on the market efficiently trading off the gains from moving workers to more productive firms against the loss of labor and decreasing returns to scale. However, the production function still has a simplified form to log second-order, as we confirm in the next lemma. We will use $f \in \{p, c\}$ to denote firm conduct, with p denoting perfect competition and c denoting Cournot.

Lemma 2. *For firm conduct $f \in \{p, c\}$, the regional production function is $Y_i^f(\ell, m) = z_i m^\eta \ell^{1-\eta} \tilde{\Phi}^f(m)$, where $z_i \equiv \mathbb{E}[z_{isn}^{1/\eta}]^\eta$. Furthermore, $\tilde{\Phi}^c(m) \leq \tilde{\Phi}^p(m)$.*

Generically, $\tilde{\Phi}^c(m) < \tilde{\Phi}^p(m)$ because when firms are imperfectly competitive, misallocation due to varying wage markdowns decreases the productivity of the market relative to the efficient benchmark.

Including imperfect competition increases the agglomeration benefits of adding more firms. The reason is that with imperfect competition, larger locations have two advantages: it is easier for a theoretical planner to move workers to their most productive use, and the equilibrium gets closer to the planner's solution. In the limit with an infinite number of firms, firms do not vary their wage markdown so that workers move to their efficient use. By contrast, in small locations, firms do not raise their wage as much as a planner would like in response to a good productivity shock, so firms expand too little. Thus, Proposition 2 only becomes stronger.

In contrast, the optimal policy changes with imperfect competition. The first-best policy would exactly replicate the perfect competition benchmark with firm- and state-dependent wage subsidies that exactly undo the wage markdown. Throughout this paper, we will assume that this policy is infeasible. Instead, we assume that the planner can give place-based transfers to workers and place-based entry subsidies to firms. Therefore, conditional on the firms and workers in a location, the planner takes as given how labor markets operate. In math, this implies that the planner will take as given the functions $\tilde{\Phi}^f(m)$.

Then, to explore optimal policy, we need to know what workers and firms are paid in equilibrium, and how that differs from their marginal contribution to production. With perfect competition, workers are paid their marginal product, so the optimal policy features firm entry subsidies, as firms cannot get paid their marginal product. If firms exploit their market power, there are wage markdowns, and workers are not paid their marginal product. Instead, their earnings are distorted downwards. The following lemma summarizes how much.

Lemma 3. *For firms competing à la Cournot, there exists a function $\Psi^c(m) \leq 0$ such that total wage compensation in location i can be written,*

$$w_i \ell_i = (1-\eta) z_i(m_i)^{\eta} (\ell_i)^{1-\eta} \left(\tilde{\Phi}^c(m_i) + \Psi^c(m_i) \right).$$

We derive $\Psi^c(m)$ in Appendix A and use it when considering our mechanism quantitatively in Section 6. Comparing the marginal product of labor and firms to their payments then implies our next result.

Proposition 4. *The optimal policy features an earnings subsidy in location i at rate τ_i^w and a firm entry subsidy (or tax) at rate τ_i^π, where*

$$1 + \tau_i^w = \frac{1}{1 + \frac{\Psi^f(m_i)}{\tilde{\Phi}^f(m_i)}}; \qquad 1 + \tau_i^\pi = \frac{1 + \frac{1}{\eta} \frac{\partial \log \tilde{\Phi}^f(m_i)}{\partial \log m}}{1 - \frac{1-\eta}{\eta} \frac{\Psi^f(m_i)}{\tilde{\Phi}^f(m_i)}}. \tag{13}$$

Without any more information, we know that the optimal policy features a subsidy on wages, $\tau_i^w > 0$. Those subsidies will undo the average markdown workers face in location i. The sign of τ_i^π, on the other hand, is ambiguous. As we showed in Proposition 3, granularity implies that there is under-entry. This force is represented in the numerator of equation (13) as $\frac{\partial \log \tilde{\Phi}^f(m)}{\partial \log m} > 0$ and pushes for the optimal subsidy to be positive. However, with imperfect competition, there is another force in the denominator. As discussed by Mankiw and Whinston (1986), when firms are imperfectly competitive, there is a tendency for too many firms to enter. In our model, this arises because firms mark down wages, biasing profits above their competitive levels, increasing the incentive to enter. The quantitative importance of this force is summarized by the size of the wage markdown, $\frac{\Psi^f(m)}{\tilde{\Phi}^f(m)}$. We return to how these forces interact quantitatively in Section 6.

4 Estimation of Granular Agglomeration

In Section 3, we presented a quantitative model of economic geography with granular firms. In this section, we estimate the model in order to quantify how important the granular mechanism is for agglomeration in Section 6. We focus on manufacturing sectors in Japan, where high-quality firm-level data across years enable us to examine the extent to which firms experience idiosyncratic shocks.

4.1 Data

Data Sources. Our primary data source is the Census of Manufacture (CoM) in Japan, conducted annually by the Ministry of Economy, Trade, and Industry (METI). The census covers all manufacturing establishments in years ending with 0, 3, 5, or 8, and those with at least four employees in other years. The CoM was not conducted in 2012 or 2016, which would have covered 2011 and 2015, respectively; instead, the Economic Census for Business Activity (ECBA), conducted by METI and the Ministry of Internal Affairs and Communications, provides data for 2011 and 2015.[14] We use the ECBA as a substitute for the CoM in 2011 and 2015.

We use data from 2002–2019, as product classification changed discontinuously in 2002. For establishment-level analysis, we restrict the sample to establishments with at least 10 employees and those surveyed in at least 10 of the 18 years (2002–2019).

These Japanese data have two advantages. First, we observe panels of all establishments with at least 4 employees. This feature allows us to compute a variety of volatility measures at the establishment level as well as commuting zone and sector-level variables.[15] Second, we observe yearly shipment values by detailed, 6-digit product categories for each establishment-year. This enables us to construct establishment-level exposure to product-level demand shocks at an annual frequency to test some of the predictions.

Mapping from Model to Data. We interpret region i as a commuting zone and denote by $\mathcal{I}$ the set of 225 commuting zones in Japan.[16] We map sectors $s \in \mathcal{S}$ onto 148 manufacturing industries at the 3-digit level.[17] Each establishment in the data is treated as an independent firm, as multi-establishment firms are not modeled. We refer to each establishment as a "firm" throughout. We interpret the short run as 1 year, which is enough time for workers to move between firms but not across commuting zones.

4.2 Labor Supply

Short Run Labor Supply Elasticity across Firms. We first estimate a short-run labor supply elasticity across firms, κ. As we show in (28) in the Appendix, the labor supply equation in period 2 is given by

[14] The ECBA covers all establishments, including non-manufacturing ones, but we focus on manufacturing to maintain consistency with the CoM.

[15] One further advantage, compared to the US LBD data, is that we can separately identify single establishments within each of the 47 prefectures.

[16] To construct time-consistent commuting zones, we follow Kondo (2023) to convert municipalities into time-consistent groups. Japan has 1,724 municipalities as of June 2023, including 6 in the Northern Territories, which we drop because the CoM does not cover them. We then use the converter in Adachi et al. (2020) to map these groups into commuting zones and retain those with at least 10 manufacturing establishments in 2019.

[17] We use a RIETI crosswalk to convert all categories into 2011 codes. In theory, sectors form a continuum, but we use 148 finite sectors because workers move freely within these broad groups, which best represent labor markets. This abstraction omits sectoral granularity, underestimating individual firm influence.

$$\log \ell_{isn}(\omega) - \log \ell_{isn} = \kappa \log w_{isn}(\omega) - (\kappa - \nu) \log w_{is}(\omega) - \nu \log w_i(\omega) + \tilde{\epsilon}^w_{isn}(\omega). \tag{14}$$

We interpret the short run of period 2 as one year. Therefore, to estimate κ, we take a one-year change of these variables and get the following equation:

$$\Delta \log \ell_{isnt} = \kappa \Delta \log w_{isnt} + \gamma_{ist} + \tilde{\epsilon}^w_{isnt}, \tag{15}$$

where γ_{ist} is a market-time fixed effect, which allows us to focus on variations across firms within location-sector-year.

The key threat to the identification of κ is that changes in wages of firm n in location i and sector s might be correlated with the changes in workers' taste for firm n. To address this concern, we instrument $\Delta \log w_{isnt}$ with a shift-share IV, Δd_{isnt}, constructed as follows.

$$\Delta d_{isnt} \equiv \sum_p \overline{s^p_{isn}} \cdot \Delta d^{\text{national}}_{pt}, \tag{16}$$

where $\overline{s^p_{isn}}$ is a time-invariant share of product p in total shipment from firm n. We take the average over our sample periods between 2002 and 2019. $\Delta d^{\text{national}}_{pt}$ is a one-year log change in the shipment of product p at a national level. We interpret this shift in demand as a shock to the revenue productivity of the firm, as that shift increases the effective price of the firm's output. As we have normalized the price of each good to one, this shift in price is loaded on the effective productivity $a_{isn}(\omega)$ in the model.

The results are in Table 1. All columns include market-time fixed effects, the sum of the shares ('incomplete shares'), and the interaction of the sum and time fixed effects as covariates. Columns (2), (3), and (4) include firm-level lagged log employment growth as an additional covariate. Column (3) drops the top 10 products (units of the shifts) in terms of the absolute values of Rotemberg weights, which we present in Table A2. Column (4) uses lagged shipment share, instead of the time-invariant average shipment share, $\overline{s^p_{isn}}$, to construct the shift-share IV in (16). Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses.

The estimates across columns are not statistically different, and we take the estimate in Column (1), $\hat{\kappa} = 1.62$, as our baseline estimate. This is in line with many of the estimates in the literature. Caldwell et al. (2026) review 91 studies that estimate firm-level labor supply elasticity and report that the median is 2.14. Among the 91 studies, Lamadon et al. (2022) estimate this elasticity to be 6.52 in the US, while Felix (2026) finds $\hat{\kappa} = 1.01$ in Brazil.

Short Run Labor Supply Elasticity across Markets. Next, we estimate a short-run labor supply elasticity across markets, ν. The labor supply to sector s in period 2 can be written,

$$\log \ell_{is}(\omega) - \log \ell_{is} = \nu \log w_{is}(\omega) - \nu \log w_i(\omega) + \varepsilon^w_{is}(\omega), \tag{17}$$

Table 1: Estimation of Short-run, Across-Firm Labor Supply Elasticity

Dep. Var.: Log Employment Growth				
	(1)	(2)	(3)	(4)
Log Wage Growth	1.62	2.00	1.84	1.98
	(0.70)	(0.95)	(0.80)	(0.98)
LLM-Year FE	✓	✓	✓	✓
Control Lagged Emp Growth		✓	✓	✓
Drop Top 10 Rotemberg			✓	
Use Lagged Share				✓
1st-stage F Stat.	45.66	32.26	28.08	27.70
Num of Unique LLMs	9,379	9,246	9,222	9,246
Num of Unique Products (Shifts)	1,829	1,829	1,819	1,829
Num of Unique Firms	114,399	112,733	111,174	112,733
Num of Years	17	16	16	16
Shift-Year Obs.	30,693	28,797	28,637	28,793
Firm-Year Obs.	1,340,985	1,205,067	1,180,337	1,205,067

Note: This table shows the estimates of short-run labor supply elasticity across firms within markets, κ, following (15). They include market-time fixed effects, the sum of the shares, and the interaction of the sum and time fixed effects as covariates. Columns (2), (3), and (4) include firm-level lagged log employment growth as an additional covariate. Column (3) drops the top 10 products (units of the shifts) in terms of the absolute values of Rotemberg weights, which we present in Table A2. Column (4) uses lagged shipment share instead of the time-invariant average shipment share to construct the shift-share IV in (16). Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses.

where

$$w_{is}(\omega) \equiv \left[\sum_{n \in \mathcal{N}_{is}} \frac{\ell_{isn}}{\ell_{is}} w_{isn}(\omega)^{1+\kappa} \right]^{\frac{1}{1+\kappa}}.$$

We then estimate ν following a very similar procedure to how we estimate κ. Taking a one-year difference of these variables, we get the following equation:

$$\Delta \log \ell_{ist} = \nu \Delta \log w_{ist} + \gamma_{it} + \gamma_{st} + \varepsilon^{w}_{ist}, \tag{18}$$

where γ_{it} is a location-time fixed effect, and γ_{st} is a sector-time fixed effect. To get theory-consistent measures of the market-level labor and market-level wages, we use the estimate $\hat{\kappa} = 1.62$ to construct the variables

$$\ell_{ist} = \left[\sum_{n \in \mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \right)^{-\frac{1}{\hat{\kappa}}} \ell_{isnt}^{\frac{1+\hat{\kappa}}{\hat{\kappa}}} \right]^{\frac{\hat{\kappa}}{1+\hat{\kappa}}}, \quad w_{ist} = \left[\sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} w_{isnt}^{1+\hat{\kappa}} \right]^{\frac{1}{1+\hat{\kappa}}}.$$

We then use equation (18) to estimate ν. Just as in the case of κ, we instrument for wages to avoid endogeneity issues. We construct market-wide versions of the same instrument used to estimate κ,

$$\Delta d_{ist} \equiv \sum_{n} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \cdot \Delta d_{isnt}, \tag{19}$$

Table 2: Estimation of Short-run, Across-Market Labor Supply Elasticity

Dep. Var.: Log LLM Employment Growth				
	(1)	(2)	(3)	(4)
Log LLM Wage Growth	0.80	0.84	0.90	0.98
	(0.13)	(0.13)	(0.14)	(0.13)
Sector-Year FE	✓	✓	✓	✓
CZ-Year FE	✓	✓	✓	✓
Control Lagged Emp Growth		✓	✓	✓
Use Lagged Share			✓	
Weighted				✓
1st-stage F Stat.	455.59	475.56	401.42	1768.00
Num of Unique LLM	12,501	12,063	12,063	12,063
Num of Years	17	16	16	16
LLM-Year Obs.	174,864	159,145	159,145	159,145

Note: This table shows the estimates of short-run labor supply elasticity across markets, ν, following (18). The unit of observation is the local labor market (LLM)-year level. LLMs are defined as pairs of 3-digit sectors and commuting zones. All columns include sector-time fixed effects, cz-time fixed effects, the sum of the shares ('incomplete share'), and the interaction of the sum and time fixed effects as covariates. Columns (2), (3), and (4) add lagged employment growth as a new covariate. Column (3) used lagged shipment share when constructing the firm-level shift share IV in (16). Column (4) weights each local labor market using the median total employment over time. Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses.

where Δd_{isnt} is the shift-share IV constructed in (16), and the weight $\overline{\ell}_{isn}/\overline{\ell}_{is}$ is measured as the median, across the sample years, of the payroll share of firm n in the labor market of location i and sector s. We use the same weights to construct ℓ_{ist} and w_{ist} above. In essence, Δd_{ist} is a weighted average of firm-level demand shocks at the market level.

The results are in Table 2. All columns include sector-time fixed effects, cz-time fixed effects, the sum of the shares ('incomplete share'), and the interaction of the sum and time fixed effects as covariates. Columns (2), (3), and (4) add lagged employment growth as a new covariate. Column (3) uses the lagged shipment share when constructing the firm-level shift share IV in (16). Column (4) weights each local labor market using the median total employment over time. Exposure-robust standard errors are in parentheses.

The estimates across the specifications are similar, and we take the estimate in Column (1), $\hat{\nu} = 0.80$, as our baseline estimate. In the US setting, Lamadon et al. (2022) estimate this elasticity to be 4.57, while Felix (2026) finds $\hat{\nu} = 0.69$ in Brazil.

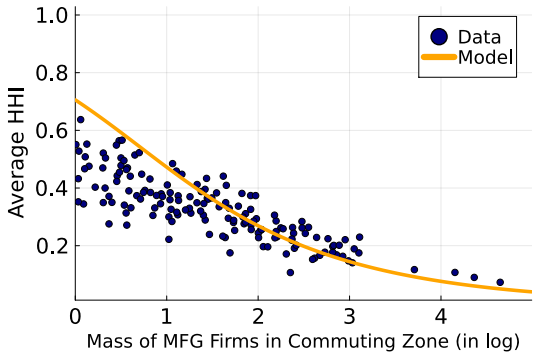
4.3 Labor Demand

Returns to scale. We set $\eta = 0.18$ to be the average firm profit share in 2019 using Financial Statements Statistics of Corporations (FSSC).[18]

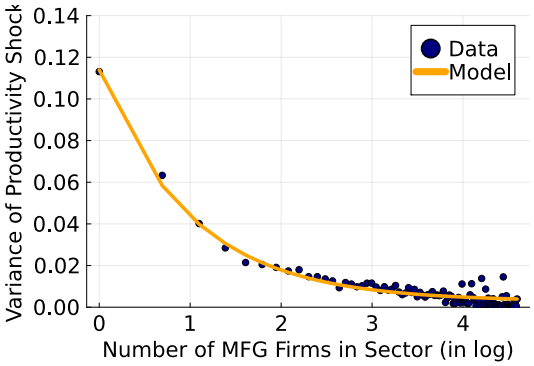
[18]Our baseline model does not have capital, so all money not paid out in profits goes to workers. In Appendix C.4, we present an extension of the model with freely traded capital and show that it is equivalent to our baseline model

Figure 2: Supply Side - Model Fit

(a) λ fit



(b) σ^N and σ^S fits



Note: The figures show the fits of the model to the moments we target. The left panel shows the average HHI across sectors against the natural logarithm of the mass of manufacturing firms in each commuting zone. The right panel shows the variance of the average productivity shock to sectors with N manufacturing firms, defined by (21), against the natural logarithm of the number of manufacturing firms per sector. In both panels, the blue dots show the data and the orange line shows the model-implied relationship.

Ex-Ante Productivity Distribution. We could estimate the shape parameter of the ex-ante productivity distribution λ in a couple of different ways. However, Equation (11) suggests that the key moment determining the strength of the externality is the average HHI across sectors within a location. Therefore, we estimate λ to match the average HHI across sectors for each location i, $\int_0^1 \frac{\bar{\ell}_{is}}{\bar{\ell}_i} HHI_{is} ds$, with a quadratic loss function. In the data, we measure HHI_{is} using payroll shares. Intuitively, a higher λ implies that the firm distribution is less thick-tailed and average HHI will decrease quickly with the mass of firms in a location i. A lower λ implies that the HHI would decrease much more slowly as a few large firms will dominate most markets. The fit is depicted in Figure 2.

Ex-Post Productivity Distribution. To estimate the distribution of ex-post productivity shocks, we look at the distribution of the average shock to a sectoral labor market as a function of the number of firms in that market. Recall that production is $y_{isn}(\omega) = z_{isn} a_{isn}(\omega) \ell_{isn}(\omega)^{1-\eta}$. As we interpret the time scale of period 2 as one year, we take first differences to get

$$\Delta \log a_{isnt} = \Delta \log y_{isnt} - (1-\eta) \Delta \log \ell_{isnt}. \tag{20}$$

Using our estimate of η, we can estimate the productivity shock to firm n from equation (20). We then create a measure of the variance of the average productivity shock to sectoral labor markets

with an adjusted η.

Table 3: Summary of Estimated and Calibrated Parameters

Description	Parameter	Value	Source
A. Labor Supply			
Short run labor elasticity across firms	κ	1.62	Estimated (CoM)
Short run labor elasticity across markets	ν	0.80	Estimated (CoM)
B. Labor Demand			
Returns to scale	η	0.18	Profit Share (Data, FSSC)
Ex-ante firm prod. tail	λ	9.54	Estimated SMM (CoM)
Variance of sector shocks	σ_S^2	0.003	Estimated GMM (CoM)
Variance of idiosyncratic shocks	σ_N^2	0.11	Estimated GMM (CoM)
C. Economic Geography Parameters			
Migration elasticity	θ	3	Redding (2016)
Avg. Productivity, Amenity, Entry Cost	ζ_i, $\overline{u}_i$, ψ_i		Exact hat algebra (2019)

Note: This table summarizes where the key parameters of the model come from. See the discussion in Section 4 for a more in-depth description.

with N firms in sector s, $\sigma_s^2(N)$, defined as,

$$\sigma_s^2(N) = \text{Var}\left(\frac{\sum_{n\in\mathcal{N}_{is}} \Delta \log a_{isnt}}{N}\middle| N_{is} = N\right). \tag{21}$$

We then estimate σ_{Ss}^2 and σ_{Ns}^2 to match $\sigma_s^2(N)$ for $N = \{1, \ldots, 99\}$. We then average σ_{Ss}^2 and σ_{Ns}^2 across sectors s, weighted by national employment shares to get σ_S^2 and σ_N^2. We show the average fit in Figure 2. We find that σ_S^2 is very small. This comes from the fact that the average productivity shocks in large sectoral labor markets are very small, suggesting that sector-wide shocks are relatively unimportant on average.

4.4 Economic Geography Parameters

We take the migration elasticity $\theta = 3$ from the literature (Redding, 2016). Our key results on the size of agglomeration benefits and the optimal place-based policies do not depend on this parameter. It matters only for our results on how the population responds to the optimal policy change. However, similar migration elasticity estimates have been found in the United States (Hornbeck and Moretti, 2024) and Indonesia (Bryan and Morten, 2019). That leaves parameters summarizing the average firm productivity ζ_i, amenity $\overline{u}_i$, and fixed cost of opening a firm ψ_i in each location i. We calibrate these parameters to exactly match 2019 data on the population, average wages, and number of firms in each location.

5 Validation of the Mechanism

Before we show the quantitative importance of the mechanism, we present some reduced-form evidence of the mechanism and demonstrate that it is consistent with the model we have

just quantified.

Our reduced-form evidence focuses on two key predictions intimately tied to the mechanism. First, we provide evidence that firms are subject to idiosyncratic shocks that "average out" in larger markets by looking at the variance of the wage bill across time in labor markets of different sizes. Second, we provide evidence that firms in larger markets expand their employment more than firms in small markets in response to positive productivity shocks and shrink more in response to negative shocks. We do this with cross-sectional and quasi-experimental evidence. For cross-sectional evidence, we show that firms in larger markets have a higher variance of log employment across time. We then do a quasi-experimental test of our mechanism. We construct productivity shocks and show that for similarly situated firms, firms that are small relative to their labor market expand more in response to the same shock.

5.1 Shocks Average Out in Large Markets

Shocks to firm revenue are passed through to wages paid to workers. Thus, we can use the total payments to workers in a sectoral labor market as a measure of the shocks to the market.

If our theory is correct, firms are subject to different shocks, so the variance of the log wage bill at the sectoral market should be lower for markets with more firms. In math, workers are paid their marginal product of labor so that $w_{isn}(\omega) = (1-\eta)z_{isn}a_{isn}(\omega)\ell_{isn}(\omega)^{-\eta}$. And the short-run labor supply is given by (14). Some straightforward algebra combining these equations then implies our first empirical prediction.

Proposition 5. *If $\sigma_N^2 > 0$, the variance of the log wage bill of a sectoral labor market decreases in the number of firms. In particular,*

$$\text{Var}\left(\log\left(\sum_{n\in\mathcal{N}_{is}} w_{isn}(\omega)\ell_{isn}(\omega)\right)\right) = \left(\frac{1+\nu}{1+\eta\nu}\right)^2\left(\sigma_S^2 + HHI_{is}\sigma_N^2\right).$$

We present a binned scatter plot of the log variance of the log wage bill by sectoral local labor market against the log number of firms in Figure 3(a). To control for compositional differences across sectors and locations, we residualize both variables with respect to sector and commuting-zone fixed effects. We include the variance of the log wage bill implied by the model as well.

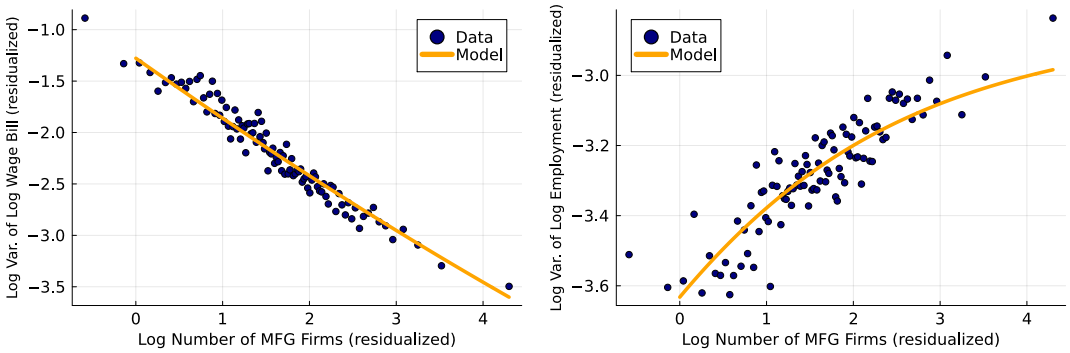
Just as predicted by the model, the variance of the log wage bill is decreasing in the number of firms. The data suggest that the rate of that decrease is slightly faster than that suggested by the model. This is likely because our model assumes that every firm has the same variance of log productivity shocks, whereas in practice, larger firms have less volatile productivity. Therefore, in the real world, the market becomes less exposed to individual firm shocks at a faster rate than that suggested by the model.[19]

[19] Estimating λ, σ_N^2, and σ_S^2 to match this bin scatter rather than how we estimate them in Section 4 does not significantly change any results.

Figure 3: Model Validation

(a) Variance of Log Wage Bill

(b) Variance of Log Employment



Note: The figures show the fits of the model to the moments we do not target. The left panel shows the variance of the log wage bill for different sectoral local labor markets. The right panel shows the variance of manufacturing-firm employment aggregated to the sectoral local labor market level. Both statistics are plotted against the natural logarithm of the number of manufacturing firms in each sector-by-commuting-zone local labor market. Before constructing the bins, both plotted variables are residualized with respect to 3-digit-sector and commuting-zone fixed effects to control for compositional differences across sectors and locations. In both panels, the blue dots show the binned data and the orange line shows the model-implied relationship. The level of the model-implied relationship (only the level, not the slope) is normalized so that the distance between the model and the data is minimized.

5.2 Firms Expand More in Larger Markets

We next turn to validating the model's key prediction for the mechanism: firms expand more in response to productivity shocks in larger markets. We provide two separate pieces of evidence for this prediction.

Cross-Sectional Evidence. We first give cross-sectional evidence for the prediction. In particular, we look at the variance of the log employment of individual firms in different-sized markets. If our theory is correct, and firms in larger markets are subject to shocks similar to those in smaller markets, then firms in larger markets should have a higher variance of log employment as they respond more to the same shocks.

In math, we can find how employment at an individual firm n responds to its own productivity shock and the shocks to every other firm to first order,

$$\Delta \log \ell_{isn}(\omega) = \frac{1}{\eta + \frac{1}{\kappa}} \left(\Delta \log a_{isn}(\omega) - \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \sum_{n' \in \mathcal{N}_{is}} \frac{\ell_{isn'}}{\ell_{is}} \Delta \log a_{isn'} \right). \tag{22}$$

Then, with some algebra, we can solve for the average variance of log employment, weighted by the average labor share. This implies our second empirical prediction.

Proposition 6. *If workers are more mobile across firms within a sector than across sectors, i.e.* $\kappa > \nu$, *and* $\sigma_N^2 > 0$, *then the weighted average variance of log employment increases with the number of firms in a*

sectoral labor market. In particular,

$$\sum_{n\in\mathcal{N}_{is}} \frac{\ell_{isn}}{\ell_{is}} \operatorname{Var}\left(\log \ell_{isn}(\omega)\right) = \left(\frac{1}{\eta+\frac{1}{\nu}}\right)^2 \sigma_S^2 + \left(\frac{1}{\eta+\frac{1}{\kappa}}\right)^2 \sigma_N^2 - \left(\frac{1}{\nu}-\frac{1}{\kappa}\right) \frac{\eta+\frac{1}{\kappa}+\eta+\frac{1}{\nu}}{\left(\eta+\frac{1}{\kappa}\right)^2\left(\eta+\frac{1}{\nu}\right)^2} HHI_{is}\sigma_N^2.$$

We plot the bin scatter for the weighted log variance of log employment for each firm against the log number of firms in Figure 3(b), applying the same residualization. We include the model-implied value as well.

Just as predicted by the model, the variance of log employment for individual firms is increasing with the number of firms. Furthermore, although we do not target it, our model matches the rate of that increase well. We even see evidence that the variance of log employment stops increasing as much for sectoral labor markets with a large number of firms as would be suggested by the model.

One thing we do not hit is the level of the variance of log employment. Our model implies a higher variance in log employment than what we see in the data. This could be for a variety of reasons. One possible reason is that the employment data measures employment in June, while firms can adjust their employment throughout the year in response to shocks.

Employment Response to Demand Shocks. We conclude with a more direct test of the mechanism by examining whether firms that hire only a small share of the sectoral labor market expand employment more in response to a revenue productivity shock.

To test this prediction, we run the regression suggested by equation (22),

$$\Delta \log \ell_{isn,t} = \beta_1 \Delta d_{isn,t} + \beta_2 \left(\frac{\ell_{isn,t}}{\ell_{is,t}}\right) + \beta_3 \left(\Delta d_{isn,t} \times \frac{\ell_{isn,t}}{\ell_{is,t}}\right) + \zeta_t + \zeta_{isn} + \varepsilon_{isn,t} \tag{23}$$

where $\Delta d_{isn,t}$ is the firm-level shock constructed as in (16) for firm n in location i and sector s. ζ_t is the time fixed effect, and ζ_{isn} is the firm fixed effect. We measure the share $\ell_{isn,t}/\ell_{is,t}$ with the payroll share of firm n in its local labor market in the base year t. Importantly, we leave out any industry-time fixed effects because our mechanism follows from the fact that an idiosyncratic shock to a large firm is also an industry-wide shock.

The shock implied by the theory is in terms of revenue productivity, while the shock $\Delta d_{isn,t}$ is in terms of total sales of products produced by the firm. Therefore, without an elasticity relating the price of the products produced by firm n to a shift in demand for the products, there are no predictions on the size of β_1 or β_3. However, the theory predicts that the ratio is given by

$$\frac{\beta_3}{\beta_1} = -\frac{\frac{1}{\nu}-\frac{1}{\kappa}}{\eta+\frac{1}{\nu}} \approx -0.441.$$

We report the results in Table 4. All columns include firm fixed effects, the sum of the shares ('incomplete shares'), its interaction with the payroll share, and the interactions of both with time

Table 4: Responses of Employment to Product-level Shocks

	Dep. Var.: Log Employment Growth			
	(1)	(2)	(3)	(4)
Shock	0.048	0.047	0.045	0.045
	(0.005)	(0.005)	(0.005)	(0.005)
Payroll Share	-0.023	-0.021	-0.020	-0.020
	(0.005)	(0.006)	(0.006)	(0.006)
Shock x Payroll Share	-0.028	-0.030	-0.029	-0.029
	(0.010)	(0.009)	(0.009)	(0.009)
Implied Ratio	-0.586	-0.634	-0.635	-0.646
95% CI	[-0.929, -0.243]	[-0.954, -0.314]	[-0.968, -0.302]	[-0.979, -0.314]
Firm FE	✓	✓	✓	✓
Year FE	✓	✓	✓	
CZ-Year FE				✓
Control Lag Emp Growth		✓	✓	✓
Drop Top 10 Rotemberg			✓	
Num of Unique LLMs	12,968	12,695	12,677	12,695
Num of Unique Firms	111,508	106,063	104,410	106,063
Num of Years	17	16	16	16
Firm-Year Obs	1,391,787	1,207,832	1,184,191	1,207,832

Note: This table shows the estimates of the impact of a firm-level demand shock on employment, following (23). All columns include firm fixed effects. All columns also include the sum of the shares ('incomplete shares') and its interaction with the payroll share, both interacted with time fixed effects, as covariates. Columns (1), (2), and (3) include year fixed effects. Column (4) includes CZ-year fixed effects. Columns (2), (3), and (4) include lagged log employment growth as a covariate. Column (3) drops the top 10 products in Rotemberg weights. Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses. The implied ratio is the ratio of the coefficient on the interaction term between the shock and the lagged payroll share within local labor markets, β_3, to the coefficient on the shock, β_1. The 95% confidence interval of that implied ratio is also reported.

fixed effects. Columns (1), (2), and (3) include year fixed effects. Column (4) includes CZ-year fixed effects. Columns (2), (3), and (4) include lagged log employment growth as a covariate. Column (3) drops the top 10 products in Rotemberg weights. Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses. The implied ratio is the ratio of the coefficient on the interaction term between the shock and the lagged payroll share within local labor markets, β_3, to the coefficient on the shock, β_1. The 95% confidence interval of that implied ratio is also reported.

First, the positive demand shock increases employment, and that effect is statistically significant across all specifications, consistent with the theory. Second, the elasticity of employment to the shock is smaller for firms with a larger payroll share within local labor markets. The estimated ratios range from -0.65 to -0.55 across specifications, all of which are close to, and not statistically different from, the theoretical prediction of -0.441. The model-implied ratio lies well within the 95 percent confidence intervals in every specification. These results indicate that the theoretical mechanism is both qualitatively and quantitatively consistent with the data.

One might be concerned that these results are driven by mechanisms other than our own. In Appendix D, we do a couple of tests to rule these out. First, we show that including industry-

location-time fixed effects makes β_3 statistically insignificant, consistent with the theory. We also estimate the labor supply elasticity κ separately for large and small locations and find no evidence that they are statistically different from each other.

6 Quantification of Granular Agglomeration

In this section, we present the results demonstrating the quantitative contribution of granularity to agglomeration. For all of the results, we present the results for the cases of perfect competition and Cournot competition. This allows us to demonstrate which results are robust to assumptions on firm conduct and which results are not. The results for the Bertrand competition are in Appendix E and are broadly consistent. Further, because our model abstracts from granularity across sectors, we focus only on those commuting zones with at least one firm per sector.

6.1 Strength of Agglomeration Benefits

As we show in Proposition 1, the strength of the granular origins of agglomeration is determined by the shape of $\tilde{\Phi}^f(m)$. Thus, we plot $\log \tilde{\Phi}^f(m)$ in Figure 4 (a) and (b).

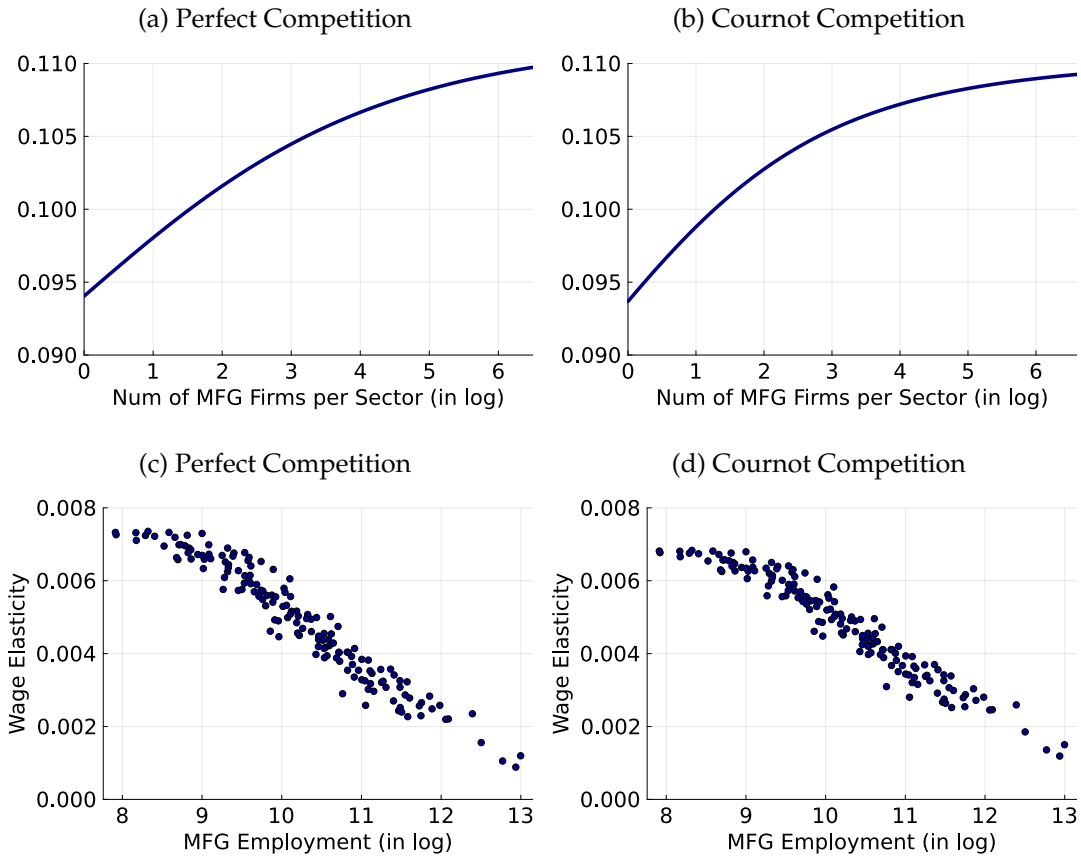
Consistent with Proposition 2, $\log \tilde{\Phi}^f(m)$ is increasing and flattens out for larger m, suggesting that the marginal benefit of increasing the number of firms in a location that is already larger is small, while locations with few firms could see large productivity benefits from increasing the number of firms. The implied $\tilde{\Phi}^f(m)$ looks very similar whether firms are competing perfectly or according to Cournot.

In total, our estimated $\tilde{\Phi}^p(m)$ implies that the largest commuting zone in Japan (Tokyo, with an average of 104.3 firms per sector) is 1.5% more productive than the smallest commuting zone (with an average of 1.01 ($\approx 149/148$) firms per sector).

In Figure 4 (c) and (d), we plot the implied elasticity of wages to population due to our mechanism for each of the commuting zones in Japan against the population. In the perfect competition case, we find that the wage elasticity gets as high as 0.0074 for the smallest commuting zone. The implied elasticity is much smaller for the large locations: Tokyo has an implied elasticity of 0.0009. If firms are competing à la Cournot, the elasticities range from 0.0012 to 0.0068. Somewhat surprisingly, Cournot competition implies a lower elasticity than perfect competition in the smallest commuting zones. This is because, as we show in the next subsection, Cournot competition features close to optimal firm entry while perfect competition implies too little entry. Therefore, with perfect competition, adding workers moves firm entry closer to the optimal level. This loading on the wedge increases the elasticity under perfect competition over that of Cournot competition.

For context, Combes et al. (2011) find that most causal estimates of the entire urban wage premium (due to all mechanisms) are the elasticities between 0.02 and 0.05 when pooling across locations of all sizes. Thus, our granular mechanism could explain as much as 37% ($= 0.0074/0.02$)

Figure 4: $\log \tilde{\Phi}^f(m)$ and Wage Elasticities



Note: Panels (a) and (b) show $\log \Phi^f(m)$ against the natural logarithm of the number of manufacturing firms per sector, defined as the total number of manufacturing establishments in a commuting zone divided by the 148 three-digit manufacturing sectors. Panels (c) and (d) show the wage elasticity against the natural logarithm of total employment at manufacturing establishments in the commuting zone in 2019. Each commuting-zone observation represents one of the Japanese commuting zones in the sample. The left column assumes perfect competition, and the right column assumes Cournot competition.

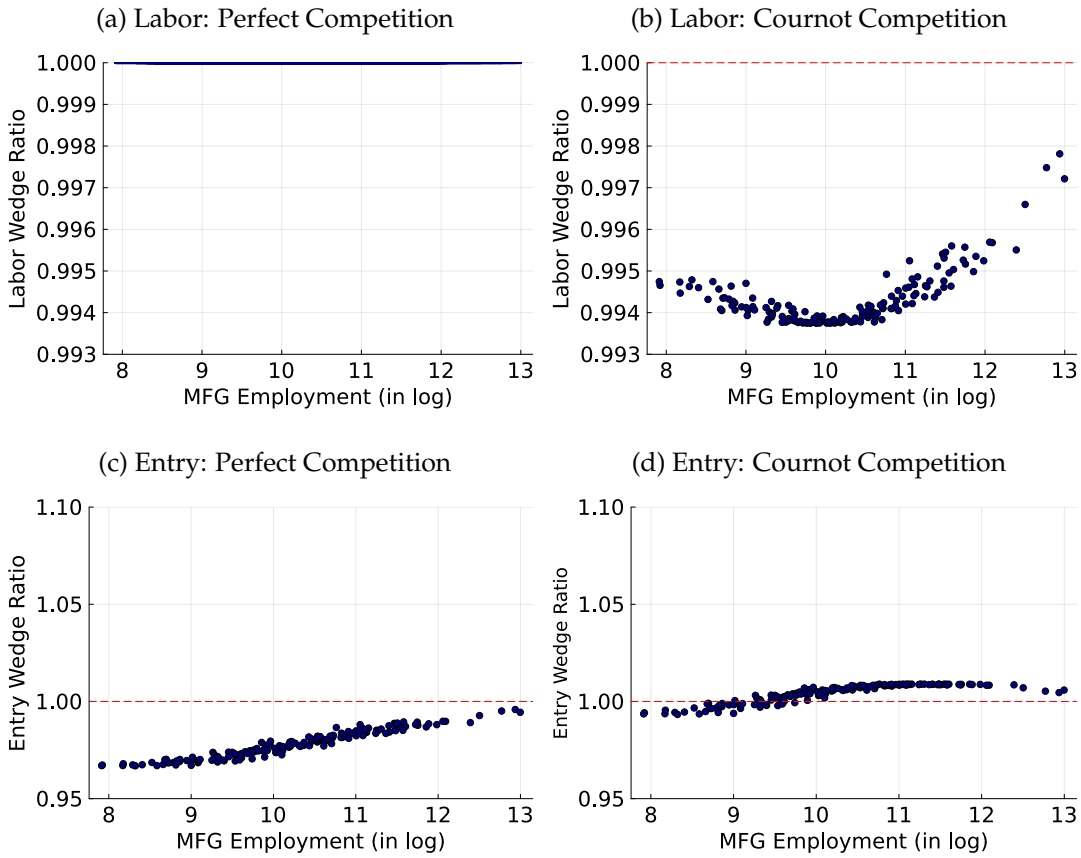
of the wage benefits of living in a large location. However, this mechanism cannot explain most of the urban wage premium.

Figure A7 shows the implied wage elasticities in alternative settings, such as the model with firm sorting, Bertrand competition in the labor market, models with capital, and the versions using labor supply elasticity estimates in the US (Lamadon et al., 2022) and those in Brazil (Felix, 2026). Except for the ones with the estimates in Brazil, the magnitudes of our mechanism are only slightly different, and the patterns across commuting zones remain unchanged.

6.2 Factor Wedges

We next turn to quantify what our model implies for the degree of wage markdowns and under-entry.

Figure 5: Factor Wedge: Labor and Firm Entry



Note: The figures show the labor wedge ratio and firm entry wedge against the natural logarithm of total employment at manufacturing establishments in each commuting zone in 2019. Each commuting-zone observation represents one of the Japanese commuting zones in the sample. The left column assumes perfect competition; the labor wedge ratio in panel (a) is one by construction. The right column assumes Cournot competition.

Labor Wedges. We start by plotting the labor wedge ratio in Figure 5. This is the ratio of average wages to the marginal product of another worker. By assumption, the perfect competition case in Figure 5(a) is 1 for all locations. Under Cournot competition, firms' market power increases following good shocks and decreases following bad shocks, which enables them to mark down wages even though workers are freely mobile in the long run. This short-run market power implies that workers face an average markdown of up to 0.63% in small and medium-sized cities. Only in larger locations like Tokyo do the wage markdowns become as small as 0.22%. The smallest commuting zones do not have the largest average markdowns. Instead, there is a local maximum around a log population of 10. That is because firms can only exploit their market power if their share of the sectoral local labor market varies. If there is only one firm in the sector, the firm only competes against other firms in the long run, when workers are freely mobile. If there are two firms in the sector, then the firm will see large swings in its own market power and can vary its markdown to take advantage of that.

Firm Entry Wedges. We next turn to the implications for firm entry. As stated in Proposition 3, granularity implies that too few firms enter in equilibrium. In Figure 5, we plot the ratio of firm profits to the marginal product of another firm for the location's output. Consistent with the proposition, Figure 5(c) shows that every firm is paid less than its marginal product if firms are perfectly competitive. In the smallest locations, firms capture 96.7% of their contribution to production, while in the largest location they capture a full 99.6%.

As we mentioned in Section 3, the efficiency of entry depends on firm conduct assumptions. By exploiting their market power, firms distort wages downward and profits upward. This can lead profits to increase above the marginal product of another firm, taking as given the conduct of the firms, conditional on firm entry. In Figure 5(d), we plot the ratio of firm profits to the marginal product of firms. We find that there is still under-entry of firms in the smallest cities. However, the over-entry forces dominate for larger cities, including Tokyo.

6.3 Implications of Optimal Policy

In the previous subsection, we showed the ratio of factor payments to the marginal product of that factor under different firm conduct assumptions. In this subsection, we consider how the equilibrium levels of population and mass of firms would change if a national government put in place the optimal place-based transfers and firm entry subsidies suggested by Proposition 4, paid for with a proportional increase in income taxes.

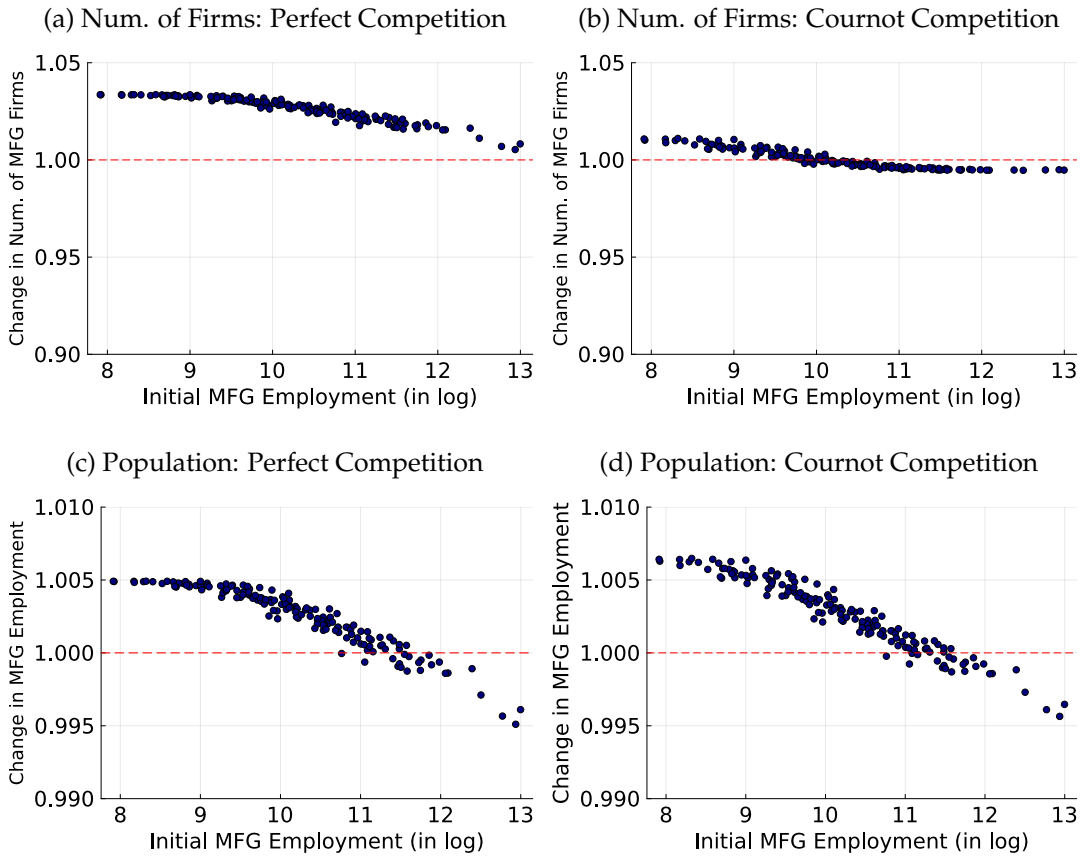
We plot how the optimal subsidies change the equilibrium number of firms and population in Figure 6. In Figure 6(a), we show that the equilibrium number of firms increases with the optimal subsidies if firms are competitive. The smallest locations see a 3.4% increase in the number of firms, while in Tokyo, the number of firms only increases by 0.5%. When firms compete à la Cournot, Figure 6(b) shows that the optimal number of firms is 101.1% of the observed number of firms in the smallest locations, while it is 99.5% in Tokyo.

Figure 6 (c) and (d) plot the effect of the optimal policy on the population. If firms compete perfectly, the optimal policy only indirectly affects where people would like to live because the optimal policy will increase the number of firms, especially in the smallest locations. Therefore, the smallest locations see a 0.5% increase in population while Tokyo sees a very small decline in population in Figure 6(c). The effects of optimal policy on population are similar when firms compete à la Cournot, as shown in Figure 6(d). Here, the optimal policy features a small subsidy on firm entry in the smallest location paired with a larger subsidy on wages. Together, these imply the optimal population is 0.6% higher than the observed population in the smallest locations.

7 Concluding Remarks

The world is granular, and local sectoral labor markets are especially so. That granularity has important implications for the geography of economic activity and optimal policy. Larger locations are more productive because the firms in those areas can expand in response to productivity

Figure 6: Changes in Number of Firms and Population



Note: The figures show the changes in the number of manufacturing firms and employment at manufacturing establishments in response to the optimal subsidies, plotted against the natural logarithm of initial total employment at manufacturing establishments in each commuting zone in 2019. Each dot represents a commuting zone in Japan. The left column shows the case of perfect competition, and the right column shows the case of Cournot competition.

shocks. A simple model of this labor market pooling mechanism in the famously illiquid labor market of Japan suggests that it could explain as much as 37% of the wage premium of large cities. Thus, while most of the urban wage premium must come from other mechanisms, granularity has important explanatory power.

In formalizing the mechanism, we were forced to ignore certain important features of real firms and labor markets. For example, we abstract from inefficient labor market adjustment to shocks such as wage rigidity and search frictions. Including these features would likely increase the social cost of being over-exposed to individual firm-level shocks as they increase the cost of adjusting to shocks more generally. Thus, this would simply increase the importance of our mechanism.

We also assume that all workers are ex-ante homogeneous, whereas many workers have specialized skills that could be used in multiple sectors. The framework we laid out can be extended to include these features. We suspect that this extension will further increase the benefits of being in a large city due to granularity, as workers with very specialized skills will want to be in very

large cities where there are enough firms that want them. The firms will then follow them as they cannot produce without those skillsets. This force will be even stronger if different skillsets are complementary in production.

On the firm side, we do not allow particular firms to direct their entry towards certain sectors or locations. Allowing for entry directed towards certain sectors could give new implications for which sectors will be overrepresented in large cities. Such a model that includes workers with specialized skills could also have important implications for coagglomeration patterns of sectors that use workers with similar skillsets.

Granularity has rich implications for the geography of economic activity, far beyond what we can explore in this paper. We think that our framework can serve as a starting point.

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